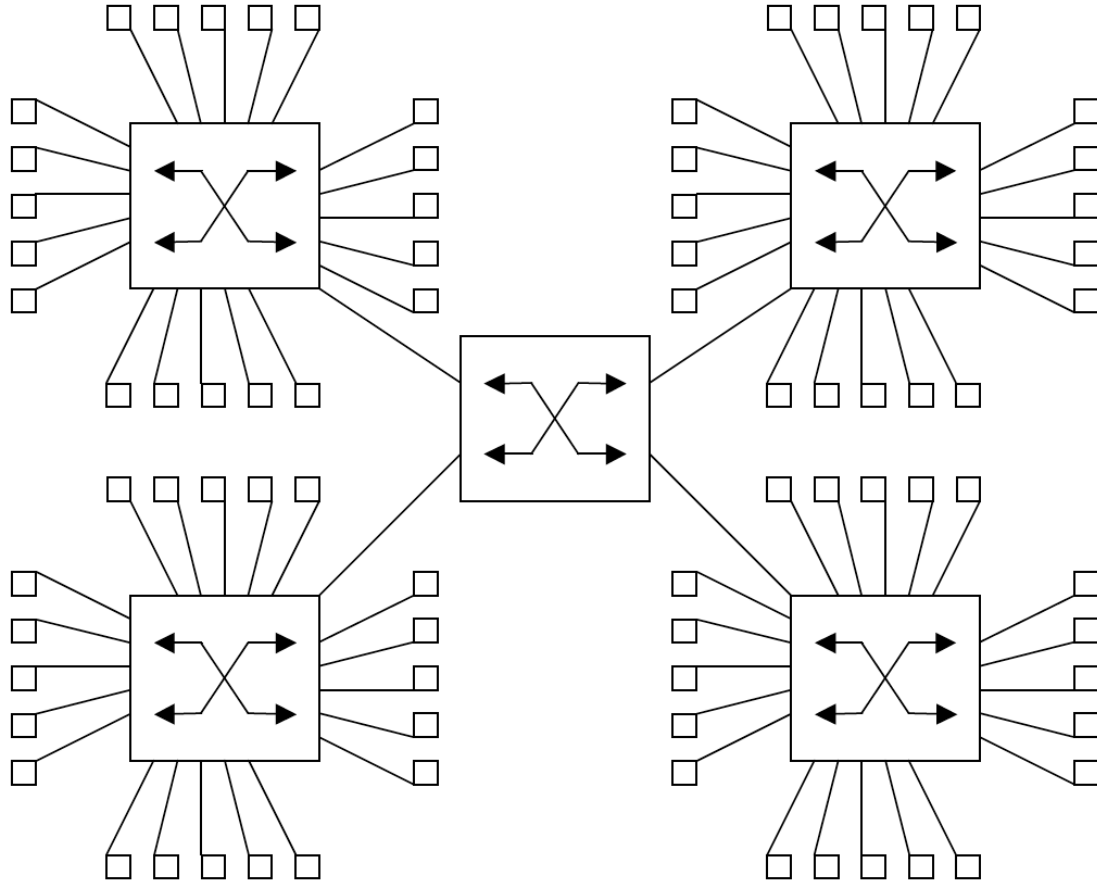


AFDX Case Study

Assume a Full-Duplex Switched Ethernet network running at 100 Mbps as illustrated in the Figure. Each peripheral switch is connected to 20 stations.



Each equipment (or End System) emits four types of frames: A, B, C and T. Frames of types A, B and C are emitted in a periodic fashion (respectively 40, 30 and 20 times per second) and have a size of 64 bytes (we will take into account the preamble and the minimal inter-frame gap, and thus arrive to an equivalent of 84 bytes). Frames of type T are aperiodic and have a size of 512 bytes (550 bytes when taking into account a header, a preamble and the minimal inter-frame gap). The minimum inter-arrival time of Frames T is 1s and the maximum jitter is 0.5s.

The aim is to evaluate the performance of this network in terms of maximum delays for any frame exchanged between any two stations. This computation will allow us to conclude on the network determinism.

Network Calculus Application :

Network model:

An End System is modeled as 4 traffic generators upper-bounded by affine envelopes, and a FCFS multiplexer.

Parameters of the affine envelopes are as followed:

$$\sigma_A = 672 \text{ bits}, \rho_A = 40 \times 672 = 26880 \text{ bps}$$

$$\sigma_B = 672 \text{ bits}, \rho_B = 30 \times 672 = 20160 \text{ bps}$$

$$\sigma_C = 672 \text{ bits}, \rho_C = 20 \times 672 = 13440 \text{ bps}$$

$$\sigma_T = 6600 \text{ bits}, \rho_T = 4400 \text{ bps}$$

These parameters are obtained with the assumption that the periodic traffic generation is perfectly regular (no jitter): the output of the generators is then a staircase function, with a step every 25 ms for the type A traffic. This function can be upper-bounded by an affine envelope whose slope is equal to the mean rate of the traffic, and whose start on the Y-axis is equal to a packet size.

Traffic of type T has a mean rate of 1 packet per second, but because of the 0.5 s jitter, this interval of 1 s might be reduced by 0.5 s, and thus 2 packets might be separated by only 0.5 s. This means that the function has a value of twice the size of a packet on the Y-axis for a value of 0.5 s. The value of σ_T is given by the equation $\sigma_T + \rho_T.t$ in this point.

1) End System's delay calculus:

- The arrival curve of the aggregated traffic generated by an End system is as follows:

$$\begin{aligned}\alpha_{\text{stat}}(t) &= \alpha_A(t) + \alpha_B(t) + \alpha_C(t) + \alpha_T(t) \\ &= 8616 + 64880.t\end{aligned}$$

- The service curve offered by an End system with a FIFO scheduling policy is simply: $\beta_{\text{stat}}(t) = 10^8.t$
- The maximal bound on traversal delay of an End system is the maximal

horizontal distance between α_{stat} and β_{stat} (application of Theorem 1)

$$D_{\text{stat}} = 8616/10^8 = 86,16 \text{ us}$$

- The output arrival curve of an End system is obtained using Theorem 2:

$$\begin{aligned}\alpha_{\text{stat}}^*(t) &= \alpha_{\text{stat}}(t + D_{\text{stat}}) \\ &= 8622 + 64880.t\end{aligned}$$

2) Peripheral switch's delay Calculus for the traffic destined to the central switch:

Only the port linked to the central switch can be studied, because studying the other output ports (linked to End Systems) would require knowledge of the traffic coming from the other switches, which we don't know yet.

- The input arrival curve of the peripheral switch is then:

$$\begin{aligned}\alpha_{\text{sp}}(t) &= 20 * \alpha_{\text{stat}}^*(t) \\ &= 20 * (8622 + 64880.t) \\ &= 172440 + 1297600.t\end{aligned}$$

- The service curve offered by the peripheral switch is a Rate-Latency curve. The latency is equivalent to the buffer traversal delay at the input port of the switch which implements the Store and forward mode and it is given by the traversal time of the largest frame T. The rate is simply the transmission capacity offered by the switch 10Mbps:

$$\beta_{\text{sp}}(t) = 10^8.(t - 44.10^{-6})$$

- The traversal delay in this case is given by Theorem 1 and it represents the maximal horizontal distance between the two curves α_{sp} et β_{sp} :

$$D_{\text{sp}} = 172440/10^8 + 44.10^{-6} = 1.768 \text{ ms}$$

- The output arrival curve of the peripheral switch is calculated thanks to Theorem 2:

$$\begin{aligned}\alpha_{\text{sp}}^*(t) &= \alpha_{\text{sp}}(t + D_{\text{sp}}) \\ &= 174735 + 1297600.t\end{aligned}$$

3) Central switch's delay calculus:

- The input arrival curve of the central switch is:

$$\begin{aligned}\alpha_{sc}(t) &= 3 * \alpha_{sp}^*(t) \\ &= 3 * (174735 + 1297600.t) \\ &= 524205 + 3892800.t\end{aligned}$$

- The service curve offered by the central switch is exactly the same than the peripheral switch:

$$\beta_{sc}(t) = 10^8 . (t - 44 * 10^{-6})$$

- The maximal bound on traversal delay of the central switch is the maximum horizontal distance between the two curves α_{sc} and β_{sc} :

$$D_{sc} = 524205 / 10^8 + 44 * 10^{-6} = 5.286 \text{ ms}$$

- The output arrival curve of the central switch is obtained thanks to Theorem 2:

$$\begin{aligned}\alpha_{sc}^*(t) &= \alpha_{sc}(t + D_{sc}) \\ &= 544783 + 3892800.t\end{aligned}$$

4) Characterization of the traffic entering an End System

We now have (finally!) knowledge of all the traffic that enters the second peripheral switch; this will allow us to characterize the traffic entering an End System.

- The arrival curve of the traffic going to an End system is:

$$\begin{aligned}\alpha_{sp2}(t) &= \alpha_{sc}^*(t) + 19 * \alpha_{stat}^*(t) \\ &= 688023 + 5125520.t\end{aligned}$$

- The service curve offered by the second peripheral switch still the same $\beta_{sp2}(t) = 10^8 . (t - 44 * 10^{-6})$

- The traversal delay of the second switch is then the horizontal distance between α_{sp2} and β_{sp2} :

$$\begin{aligned}D_{sp2} &= 688023 / 10^8 + 44 * 10^{-6} \\ &= 6.924 \text{ ms}\end{aligned}$$

5) End to end delays Calculus

In order to calculate the maximum duration required for a frame to go from one End System to another, we now just need to add all the latencies on the path.

- For a frame going to an End system linked to the same peripheral switch the delay corresponds to:

$$\begin{aligned}D_{BeB} &= D_{stat} + D_{sp2} \\ &= 7.01 \text{ ms}\end{aligned}$$

- For a frame going to an end system connected to a different peripheral switch, the delay is:

$$\begin{aligned}D_{BeB} &= D_{stat} + D_{sp} + D_{sc} + D_{sp2} \\ &= 14.064 \text{ ms}\end{aligned}$$

6) Conclusion on the network determinism

The obtained maximum end-to-end delays are smaller than the most constrained temporal deadline (25ms). Hence, we conclude that the network is deterministic from the temporal point of view.

If we have in addition memory constraints within the switches, we have to compare the maximum queue size that can be buffered in any switch (sp, sc or sp2) and the maximum offered memory to conclude if we have any risk of overflow. If yes, the network is not deterministic from the HW point of view.